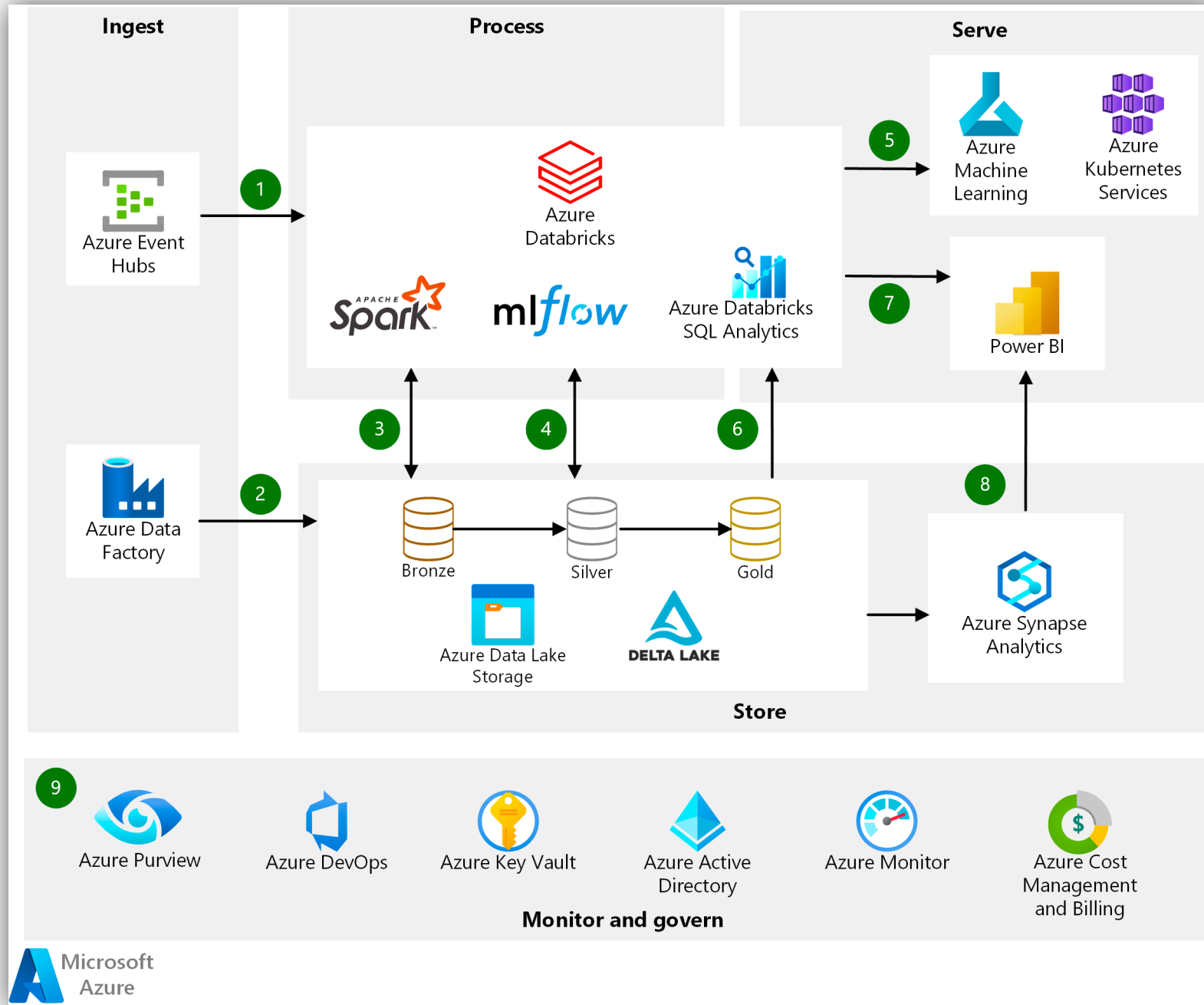


Azure Data Engineering

ADF + Databricks

Architecture

End to End Project Flow



What is Ingestion?

Data Files



Big Data Files



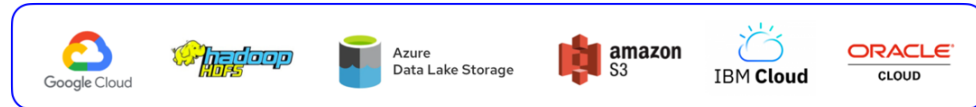
Social Media feed



IOT and Machines Generated Data



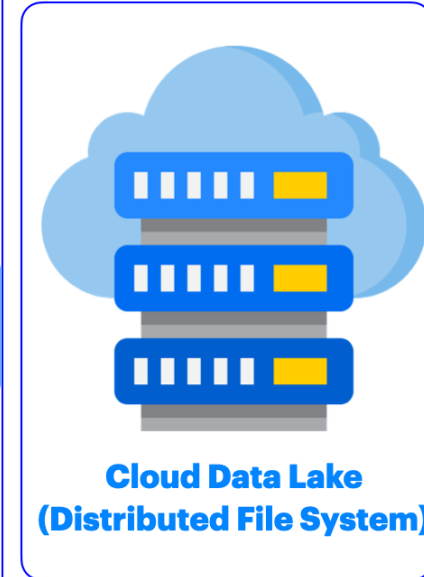
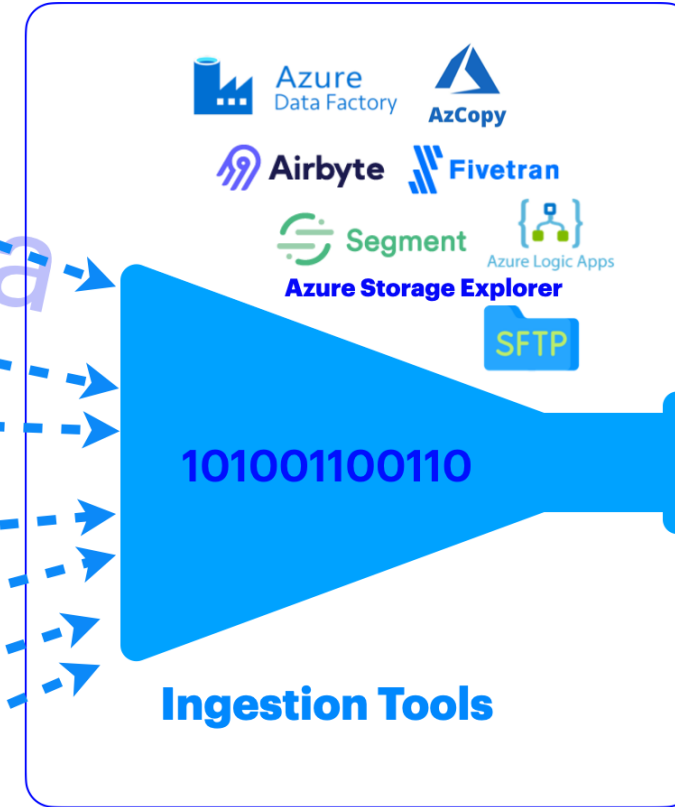
Cloud Data Lakes



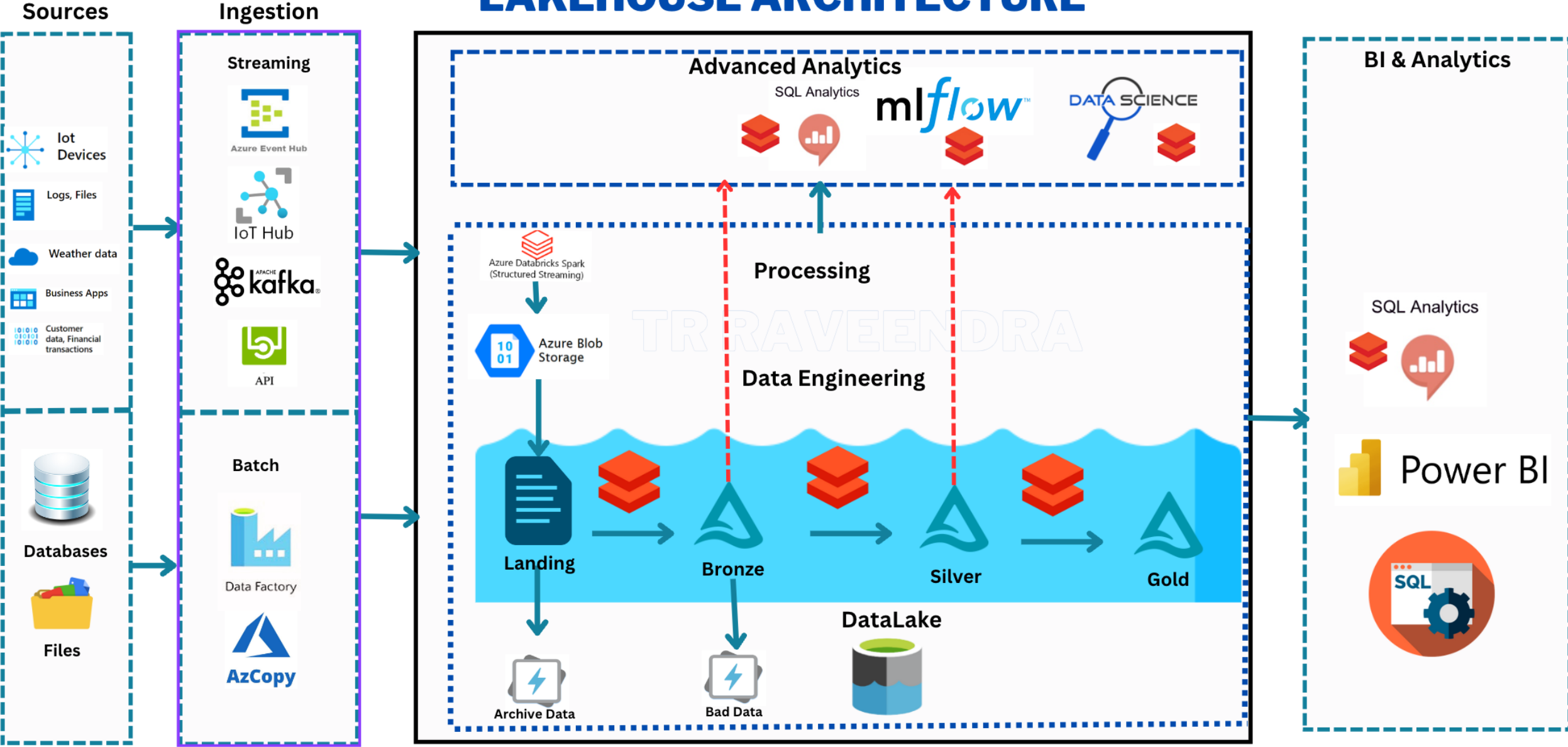
Un-Structured Data



On-Premises - DataBases



LAKEHOUSE ARCHITECTURE

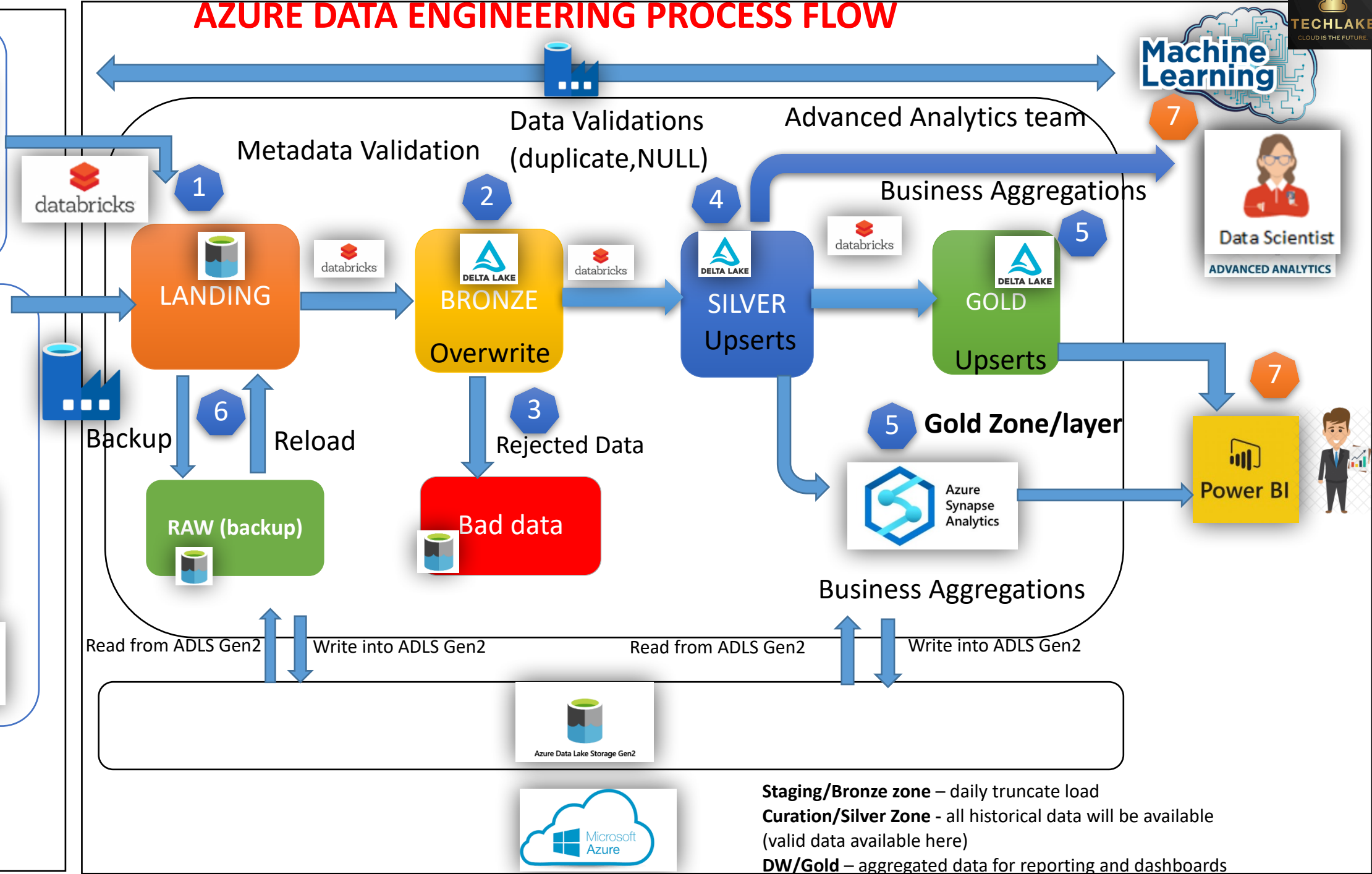
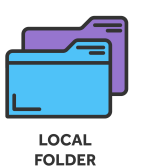


AZURE DATA ENGINEERING PROCESS FLOW

Streaming



Batch Load



COUNTRY	
Properties	
🔑 COUNTRY_ID	
🔗 COUNTRY_ISO_CODE	
🔗 COUNTRY_NAME	
🔗 COUNTRY_SUBREGION	
🔗 COUNTRY_SUBREGION_ID	
🔗 COUNTRY_REGION	
🔗 COUNTRY_REGION_ID	
🔗 COUNTRY_TOTAL	
🔗 COUNTRY_TOTAL_ID	
🔗 COUNTRY_NAME_HIST	
Navigation Properties	
🔗 CUSTOMERS	

CUSTOMER	
Properties	
🔑 CUST_ID	
🔗 CUST_FIRST_NAME	
🔗 CUST_LAST_NAME	
🔗 CUST_GENDER	
🔗 CUST_YEAR_OF_BIRTH	
🔗 CUST_MARITAL_STATUS	
🔗 CUST_STREET_ADDRESS	
🔗 CUST_POSTAL_CODE	
🔗 CUST_CITY	
🔗 CUST_CITY_ID	
🔗 CUST_STATE_PROVINCE	
🔗 CUST_STATE_PROVINCE_ID	
🔗 COUNTRY_ID	
🔗 CUST_MAIN_PHONE_NUM...	
🔗 CUST_INCOME_LEVEL	
🔗 CUST_CREDIT_LIMIT	
🔗 CUST_EMAIL	
🔗 CUST_TOTAL	
🔗 CUST_TOTAL_ID	
🔗 CUST_SRC_ID	
🔗 CUST_EFF_FROM	
🔗 CUST_EFF_TO	
🔗 CUST_VALID	
Navigation Properties	
🔗 COUNTRY	
🔗 SALES	

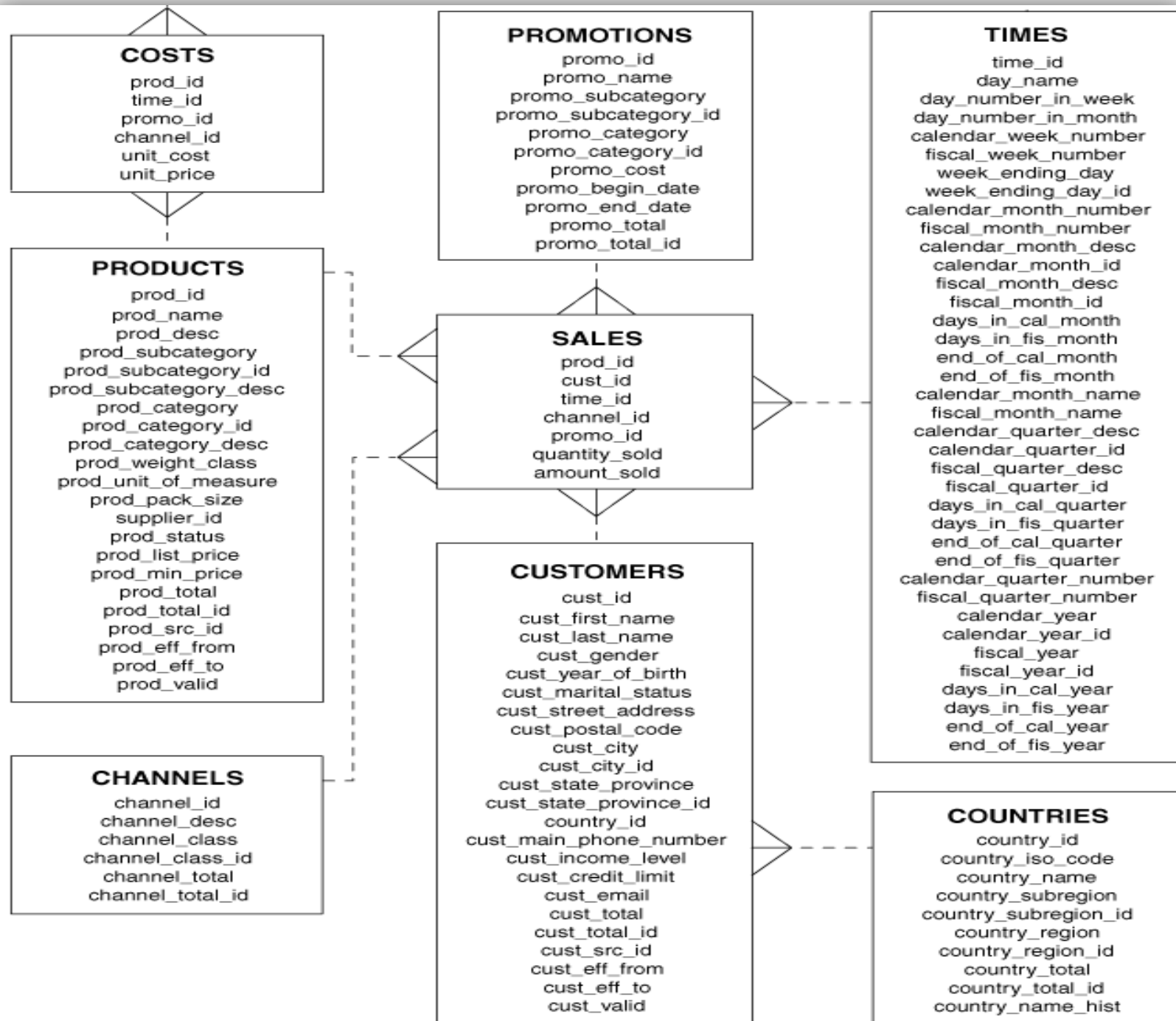
SALE	
Properties	
🔑 PROD_ID	
🔑 CUST_ID	
🔑 TIME_ID	
🔑 CHANNEL_ID	
🔑 PROMO_ID	
🔗 QUANTITY_SOLD	
🔗 AMOUNT_SOLD	
Navigation Properties	
🔗 CHANNEL	
🔗 CUSTOMER	
🔗 PRODUCT	
🔗 PROMOTION	
🔗 TIME	

PROMOTION	
Properties	
🔑 PROMO_ID	
🔗 PROMO_NAME	
🔗 PROMO_SUBCATEGORY	
🔗 PROMO_SUBCATEGORY_ID	
🔗 PROMO_CATEGORY	
🔗 PROMO_CATEGORY_ID	
🔗 PROMO_COST	
🔗 PROMO_BEGIN_DATE	
🔗 PROMO_END_DATE	
🔗 PROMO_TOTAL	
🔗 PROMO_TOTAL_ID	
Navigation Properties	
🔗 SALES	

TIME	
Properties	
🔑 TIME_ID	
🔗 DAY_NAME	
🔗 DAY_NUMBER_IN_WEEK	
🔗 DAY_NUMBER_IN_MONTH	
🔗 CALENDAR_WEEK_NUMBER	
🔗 FISCAL_WEEK_NUMBER	
🔗 WEEK_ENDING_DAY	
🔗 WEEK_ENDING_DAY_ID	
🔗 CALENDAR_MONTH_NUMBER	
🔗 FISCAL_MONTH_NUMBER	
🔗 CALENDAR_MONTH_DESC	
🔗 CALENDAR_MONTH_ID	
🔗 FISCAL_MONTH_DESC	
🔗 FISCAL_MONTH_ID	
🔗 DAYS_IN_CAL_MONTH	
🔗 DAYS_IN_FIS_MONTH	
🔗 END_OF_CAL_MONTH	
🔗 END_OF_FIS_MONTH	
🔗 CALENDAR_MONTH_NAME	
🔗 FISCAL_MONTH_NAME	
🔗 CALENDAR_QUARTER_DESC	
🔗 CALENDAR_QUARTER_ID	
🔗 FISCAL_QUARTER_DESC	
🔗 FISCAL_QUARTER_ID	
🔗 DAYS_IN_CAL_QUARTER	
🔗 DAYS_IN_FIS_QUARTER	
🔗 END_OF_CAL_QUARTER	
🔗 END_OF_FIS_QUARTER	
🔗 CALENDAR_QUARTER_NUMBER	
🔗 FISCAL_QUARTER_NUMBER	
🔗 CALENDAR_YEAR	
🔗 CALENDAR_YEAR_ID	
🔗 FISCAL_YEAR	
🔗 FISCAL_YEAR_ID	
🔗 DAYS_IN_CAL_YEAR	
🔗 DAYS_IN_FIS_YEAR	
🔗 END_OF_CAL_YEAR	
🔗 END_OF_FIS_YEAR	
Navigation Properties	
🔗 SALES	

CHANNEL	
Properties	
🔑 CHANNEL_ID	
🔗 CHANNEL_DESC	
🔗 CHANNEL_CLASS	
🔗 CHANNEL_CLASS_ID	
🔗 CHANNEL_TOTAL	
🔗 CHANNEL_TOTAL_ID	
Navigation Properties	
🔗 SALES	

PRODUCT	
Properties	
🔑 PROD_ID	
🔗 PROD_NAME	
🔗 PROD_DESC	
🔗 PROD_SUBCATEGORY	
🔗 PROD_SUBCATEGORY_ID	
🔗 PROD_SUBCATEGORY_DESC	
🔗 PROD_CATEGORY	
🔗 PROD_CATEGORY_ID	
🔗 PROD_CATEGORY_DESC	
🔗 PROD_WEIGHT_CLASS	
🔗 PROD_UNIT_OF_MEASURE	
🔗 PROD_PACK_SIZE	
🔗 SUPPLIER_ID	
🔗 PROD_STATUS	
🔗 PROD_LIST_PRICE	
🔗 PROD_MIN_PRICE	
🔗 PROD_TOTAL	
🔗 PROD_TOTAL_ID	
🔗 PROD_SRC_ID	
🔗 PROD_EFF_FROM	
🔗 PROD_EFF_TO	
🔗 PROD_VALID	
Navigation Properties	
🔗 SALES	



Integrating Azure Databricks (ADB) and Azure Data Lake Storage (ADLS) Gen2

Understanding the Integration Methods

1. Mount Points:

- **What is a Mount Point?** A mount point is a directory in the Databricks File System (DBFS) that is linked to a specific location in an external storage system like ADLS Gen2.
- **When to Use Mount Points:**
 - When you want to directly access ADLS Gen2 files from within Databricks notebooks and jobs.
 - When you need to perform file operations like read, write, and delete on ADLS Gen2 files from Databricks.

2. External Locations:

- **What is an External Location?** An external location is a reference to an external storage location (like ADLS Gen2) that can be used in Databricks queries and jobs.
- **When to Use External Locations:**
 - When you want to query data from ADLS Gen2 without mounting it to DBFS mount points.
 - External Locations comes with Unity Catalog.
 - Mount points are not Access Controlled. External Locations are Access Controlled.

Creating mount-point to access datalake in azure databricks

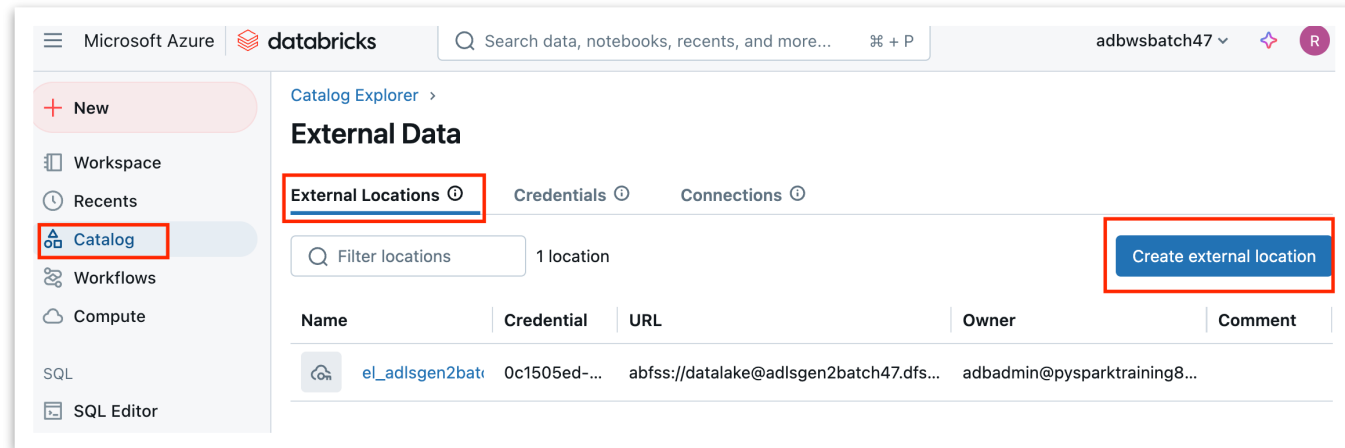
```
dbutils.fs.mount(
  source = "wasbs://<container-name>@<storage-account-name>.blob.core.windows.net",
  mount_point = "/mnt/sales",
  extra_configs = {"fs.azure.account.key.<storage-account-name>.blob.core.windows.net":dbutils.secrets.get(scope =
"<scope-name>", key = "<key-name>")})
```

7: Configure authentication for mounting

```
1 configs = {"fs.azure.account.auth.type": "OAuth",
2 "fs.azure.account.oauth.provider.type": "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider",
3 "fs.azure.account.oauth2.client.id": <client-id>,
4 "fs.azure.account.oauth2.client.secret": <service-credential>,
5 "fs.azure.account.oauth2.client.endpoint": "https://login.microsoftonline.com/<directory-id>/oauth2/token"}
```

8: Mount filesystem

```
1 dbutils.fs.mount(
2   source = "abfss://<file_system>@<storage-account-name>.dfs.core.windows.net/",
3   mount_point = "/mnt/mymount",
4   extra_configs = configs)
```



Microsoft Azure databricks

Search data, notebooks, recents, and more... + P

adbwsbatch47

New

Workspace

Recents

Catalog

Workflows

Compute

SQL

SQL Editor

Catalog Explorer >

External Data

External Locations ⓘ Credentials ⓘ Connections ⓘ

Filter locations 1 location

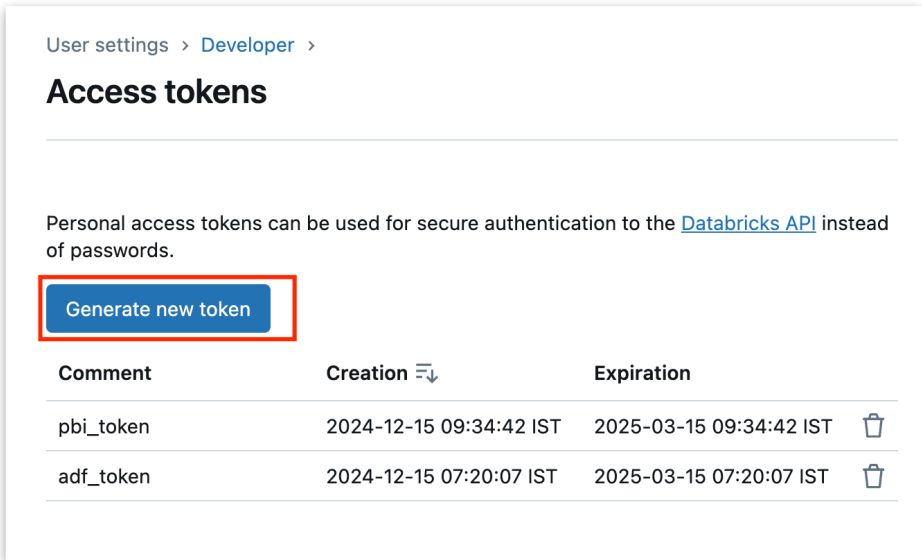
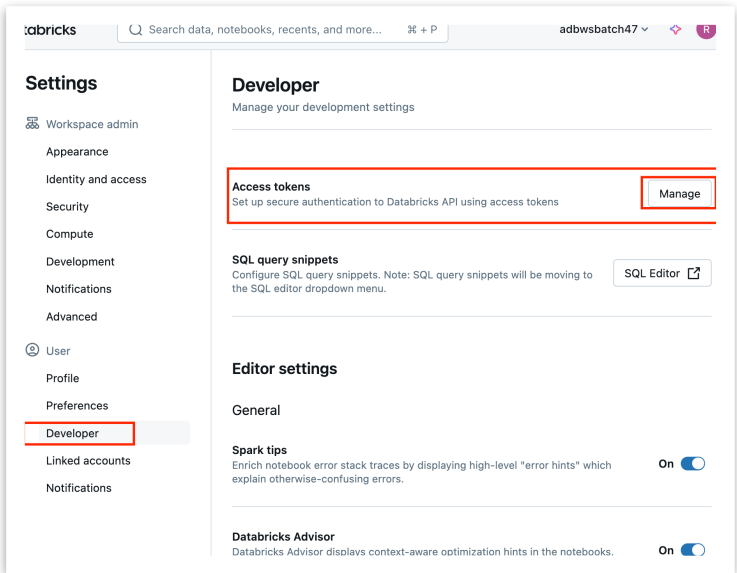
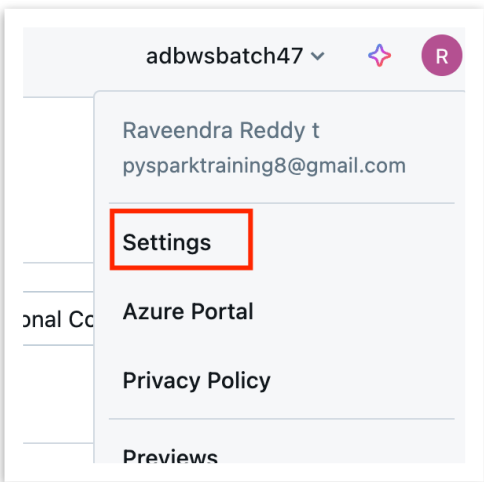
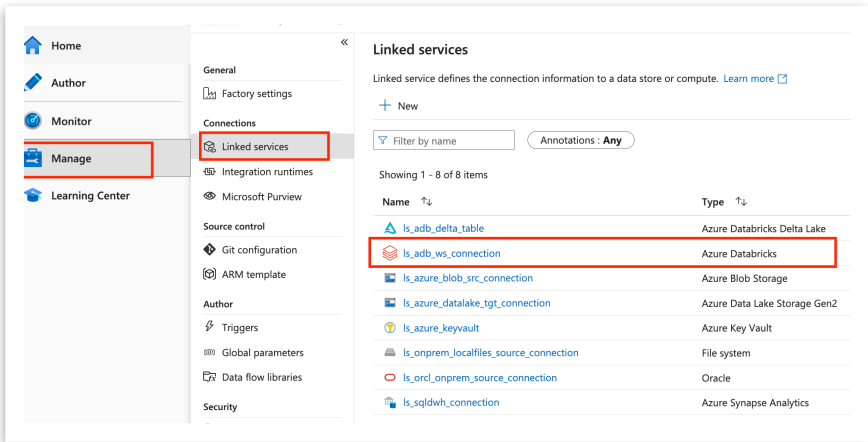
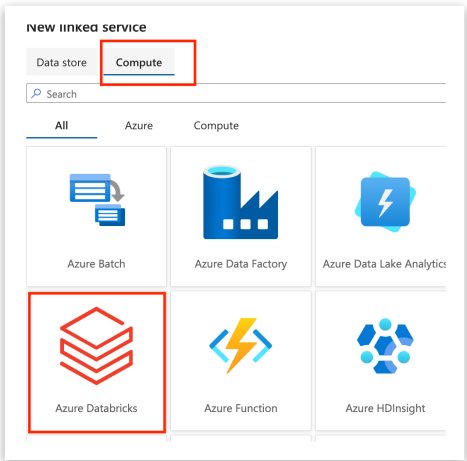
Create external location

Name	Credential	URL	Owner	Comment
e_l_adlsgen2bat	0c1505ed-...	abfss://datalake@adlsgen2batch47.dfs...	adbadmin@pysparktraining8...	

Integrating ADB with Azure Data Factory (ADF)

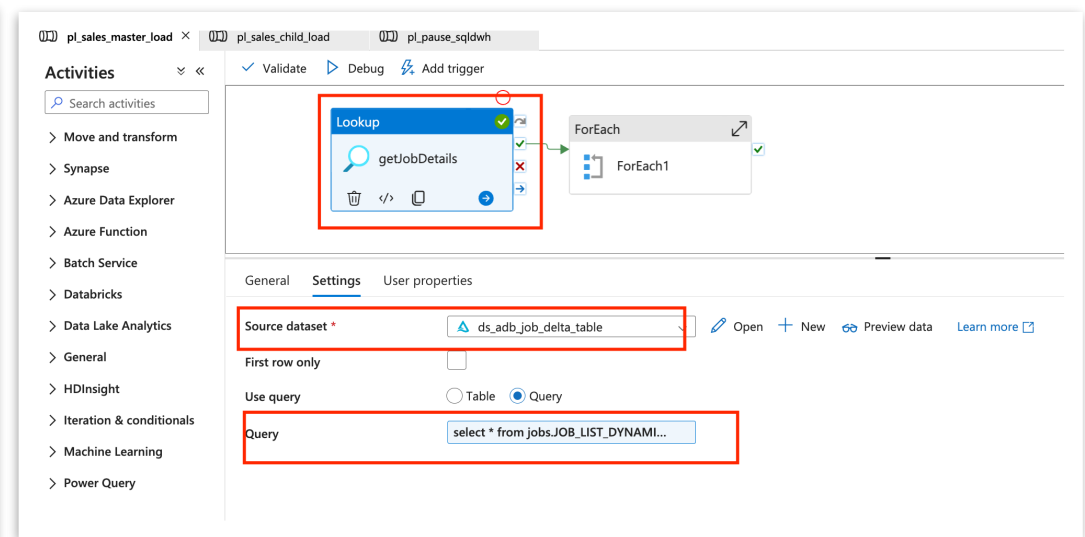
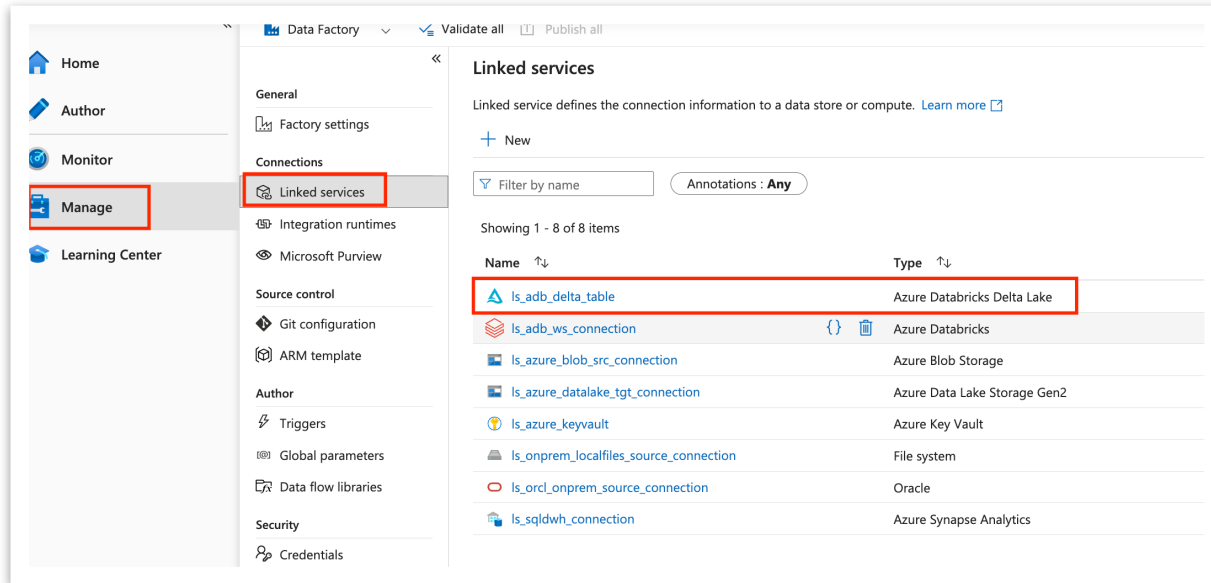
1. Authenticating ADB in ADF:

- **Token-Based Authentication:**
 1. Generate an ADB token with appropriate permissions.
 2. Store the token securely in Azure Key Vault.
 3. Create a linked service in ADF to reference the Key Vault secret containing the token.



2. Reading ADB Delta Tables in ADF:

- **Create a Databricks Delta Table Linked Service:**
 1. Configure the linked service with the necessary details, including the endpoint URL, authentication method, and Key Vault reference.
- **Create a Dataset:**
 1. Define the schema and location of the Delta table.
- **Use the Lookup Activity:**
 1. Reference the dataset to read data from the Delta table.



3. Calling Parameterized Notebooks in ADF:

- **Configure the Notebook Activity:**
 1. Set the notebook path and the necessary parameters.
 2. Map the ADF pipeline parameters to the notebook parameters under **Base Parameters**.

Activities ⌵ ⌵

Search activities

- > Move and transform
- > Synapse
- > Azure Data Explorer
- > Azure Function
- > Batch Service
- > **Databricks**
- > Data Lake Analytics
- > General
- > HDInsight
- > Iteration & conditionals
- > Machine Learning
- > Power Query

Validate Debug Add trigger

Notebook LandingToStaging

Wait wait2

Notebook StagingToCurated

Wait wai

General Azure Databricks **Settings** User properties

Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sa... Browse Open

Base parameters

+ New | Delete

Name	Value
landing_zone_file_path	@pipeline().parameters.landing_zone...
landing_zone_folder_name	@pipeline().parameters.landing_zone...
landing_zone_file_name	@pipeline().parameters.landing_zone...
landing_zone_file_type	@pipeline().parameters.landing_zone...
staging_zone_database_name	@pipeline().parameters.staging_zone...
staging_zone_table_name	@pipeline().parameters.staging_zone...

4. Sending Messages from Notebooks to ADF Activity Output:

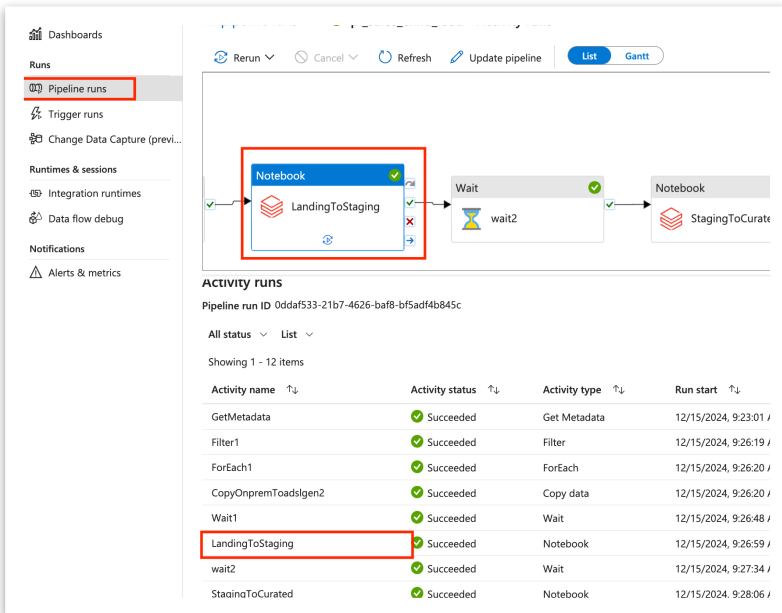
- Use `dbutils.notebook.exit(output/message)` to send a message or output to the ADF activity output.

7/18/2021 (<1s)

20

```
#dbutils.notebook.exit('SUCCESS')
#import json
dbutils.notebook.exit(json.dumps({"curationRows":curation_rowcount,"jobName":job_name,"jobId":job_id, "status":"SUCCESS"}))
```

```
{"stagingRows": 5, "curationRows": {"num_affected_rows": 5, "num_updated_rows": 5, "num_deleted_rows": 0, "num_inserted_rows": 0}, "dwhRows": {"num_affected_rows": 5, "num_updated_rows": 5, "num_deleted_rows": 0, "num_inserted_rows": 0}, "rejectedRows": 2, "jobName": "channels", "status": "Succeeded"}
```



The screenshot shows the Azure Data Factory Pipeline runs interface. The left sidebar has a red box around 'Pipeline runs'. The main area shows a pipeline diagram with a red box around the 'LandingToStaging' Notebook activity. Below the diagram is a table of activity runs.

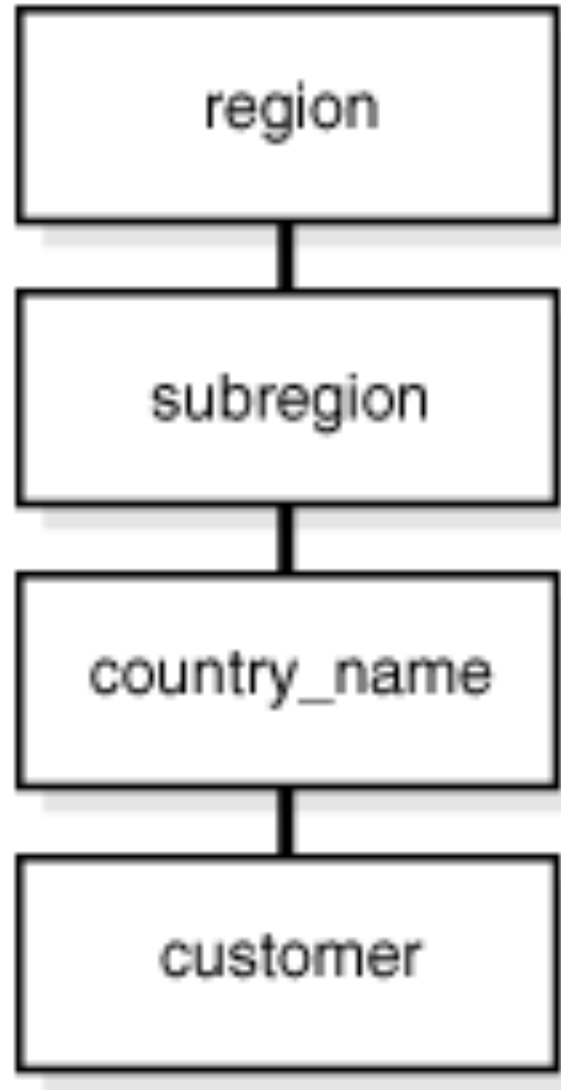
Activity name	Activity status	Activity type	Run start
GetMetadata	Succeeded	Get Metadata	12/15/2024, 9:23:01
Filter1	Succeeded	Filter	12/15/2024, 9:26:19
ForEach1	Succeeded	ForEach	12/15/2024, 9:26:20
CopyOnpremToadsgen2	Succeeded	Copy data	12/15/2024, 9:26:20
Wait1	Succeeded	Wait	12/15/2024, 9:26:48
LandingToStaging	Succeeded	Notebook	12/15/2024, 9:26:59
wait2	Succeeded	Wait	12/15/2024, 9:27:34
StagingToCurated	Succeeded	Notebook	12/15/2024, 9:28:06

Output

Copy to clipboard

```
{
  "runPageUrl": "https://adb-2218670316794790.10.azuredatabricks.net/?o=2218670316794790#job/879794882588236/run/556053580504869",
  "runOutput": {
    "landingRows": 82112,
    "stagingRows": 82112,
    "rejectedRows": 0,
    "jobName": "costs_transaction",
    "jobId": "8",
    "status": "SUCCESS"
  },
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (East US)",
  "executionDuration": 32,
  "durationInQueue": {
    "integrationRuntimeQueue": 0
  },
  "billingReference": {
    "activityType": "ExternalActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```

Typical Levels in a Dimension Hierarchy



How to Create Scope in Databricks to access Secrets from Key Vault

← → ↺ 🔒 adb-3784951373135514.14.azure.databricks.net/?o=3784951373135514#secrets/createScope

Apps What's New in Azur... Script Download Li... Azure Data Factory... Beautiful, Free peo... Today's Famous Birt... Today in I

Microsoft Azure | Databricks

HomePage / Create Secret Scope

Create Secret Scope |

A store for secrets that is identified by a name and backed by a specific store type. [Learn more](#)

Scope Name ?

batch19-adlsgen2-scope

Manage Principal ?

Creator

Azure Key Vault ?

DNS Name

https://azurekv-ravi.vault.azure.net/

Resource ID

/subscriptions/e0bf1971-b822-46ca-8e4a-72555f0a959e/resourceGroups/rg-de-dev

1/3

How to Create Scope in Databricks to access Secrets from Key Vault

azurekv-ravi | Properties

Key vault



Save



Discard changes



Refresh

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Events

Settings

- Keys
- Secrets
- Certificates
- Access policies
- Networking
- Security
- Properties**

Name	azurekv-ravi
Sku (Pricing tier)	Standard
Location	eastus
Vault URI	https://azurekv-ravi.vault.azure.net/
Resource ID	/subscriptions/e0bf1971-b822-46ca-8e4a-72555f0a959e/resourceGroups/rg-de-dev/providers
Subscription ID	e0bf1971-b822-46ca-8e4a-72555f0a959e
Subscription Name	Pay-As-You-Go
Directory ID	89556edd-0e4a-43c1-bbcb-9139ff026b1f
Directory Name	Default Directory
Soft-delete	Soft delete has been enabled on this key vault
Days to retain deleted vaults	90
Purge protection	☒ Disable purge protection (allow key vault and objects to be purged during retention period)

Cmd 9

```

1 # HDFS / DBFS / GFS / DFS / WASBS - windows azure storage blob service / AWS S3 - simple storage Service / DBFS (Databricks file system)
2 dbutils.fs.mount(
3     source = "wasbs://batch19@adlsgen2batch19.blob.core.windows.net/source/landing/sales",
4     mount_point = "/mnt/landing/sales",
5     extra_configs = {"fs.azure.account.key.adlsgen2batch19.blob.core.windows.net":dbutils.secrets.get(scope="batch19-adlsgen2-scope",key="batch19-adlsgen2-account-key")})

```

► (1) Spark Jobs

Out[2]: True

Master Pipeline

pl_master_sales_load

Activities

Search activities

Move & transform

Azure Data Explorer

Azure Function

Batch Service

Databricks

Data Lake Analytics

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Save as template

Validate

Debug

Add trigger

foreach for processing one by one job from control table and calling child pipeline

Lookup

Lookup_Getjobdetails

ForEach

ForEach_Eachjob

Activities

1 activities

Execute Pipeline

pl_alerts

Sending Email Alerts after processing

Parameters

Variables

Settings

Output

Pipeline run ID:

Name	Type	Run start	Duration	Status	Integr
------	------	-----------	----------	--------	--------

Lookup Activity

pl_master_sales_load

Activities

Search activities

- Move & transform
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Lookup

Lookup_Getjobdetails

ForEach

ForEach_Eachjob

Activities
1 activities

Execute Pipeline

pl_alerts

General **Settings** User properties

Source dataset * ds_deltalake_lookup

Use query ☐ Table ☒ Query

Query *
select * from jobs.JOB_LIST_DYNAMIC
order by job_id

First row only ☐

ForEach Activity

The screenshot shows the Azure Data Factory interface for a pipeline named 'pl_master_sales_load'. The pipeline is composed of three activities: 'Lookup_Getjobdetails' (a Lookup activity), 'ForEach_Eachjob' (a ForEach activity), and 'Execute Pipeline pl_alerts' (an Execute Pipeline activity). The 'ForEach' activity is highlighted with a red box, and its 'Settings' tab is selected. The 'Items' property is set to '@activity('Lookup_Getjobdetails').outp...'. A red arrow points from the 'Items' property to the 'Settings' tab, and another red arrow points from the 'Lookup_Getjobdetails' activity to the 'ForEach' activity.

Activities

- Move & transform
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

ForEach Settings

Sequential	☐ refer lookup activity output in foreach activity input
Batch count ⓘ	5
Items	@activity('Lookup_Getjobdetails').outp...

@activity('Lookup_Getjobdetails').output.value

Calling Child pipeline in Master Pipeline ForEach Activity to process one by one

Activities

- Move & transform
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

pl_master_sales_load > ForEach_Eachjob

Execute Pipeline

pl_child_sales_load

calling child pipeline in master pipeline inside foreach loop

Settings

Invoked pipeline * **pl_child_sales_load** Open + New

Wait on completion ☒

Parameters

specify all parameter values in pipeline and get values from foreach **@item().parameter**

Name	Type	Value
job_id	string	@item().job_id
job_name	string	@item().job_name
source_folder_name	string	@item().source_folder_name
landing_zone_file_path	string	@item().landing_zone_file_path

Chile Pipeline with Parameters

pl_master_sales_load

pl_child_sales_load

Save as template

Validate

Debug

Add trigger

Lookup

StartLogging

Get Metadata

GetMetadata_Onprem

Filter

Filter_Files

ForEach

ForEach_CopyFile

Activities
1 activities

Notebook

LandingToStaging

Wait

Wait1

Notebook

StagingToCuration

Wait

Wait

Lookup

StagingToCuration_Log

If Condition

IFDim

True
2 activities

False
No activities

Switch

SwitchFacts

Default
No activities

sales_transaction
1 activities

costs_transaction
1 activities

Parameters

Variables

Settings

Output

+ New

Delete

pl_child_sales_load

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Parameters

Variables

Settings

Output

☐	Name	Type	Default value
☐	job_id	String	Value
☐	job_name	String	Value
☐	source_folder_name	String	Value
☐	landing_zone_file_path	String	Value
☐	landing_zone_folder_name	String	Value
☐	landing_zone_file_name	String	Value
☐	staging_zone_database_name	String	Value
☐	staging_zone_table_name	String	Value
☐	staging_zone_table_pk_column	String	Value
☐	curation_zone_database_name	String	Value
☐	curation_zone_table_name	String	Value
☐	curation_zone_table_pk_column	String	Value
☐	dw_zone_database_name	String	Value
☐	dw_zone_table_name	String	Value
☐	dw_zone_table_pk_column	String	Value
☐	pyspark_schema	String	Value
☐	table_type	String	Value
☐	raw_zone_file_path	String	Value
☐	raw_zone_folder_name	String	Value
☐	raw_zone_file_name	String	Value

TECHLAKE
CLOUD IS THE FUTURE.

pl_child_sales_load ×

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Lookup

StartLogging

Get Metadata

GetMetadata_Onprem

Filter

Filter_Files

General **Settings** User properties

Source dataset * ds_deltalake_lookup Open + New Preview data

Use query ☐ Table ☒ Query

Query insert into JOBS.pipeline_log select sh...

First row only ☒

```
insert into JOBS.pipeline_log select  
sha2('@{pipeline().RunId}',256),@{pipeline().parameters.job_id},@{pipeline().DataFactory}','@{pipeline().Pipeline}','@{pipeline().RunId}','@{pipeline().TriggerType}','@{pipeline().TriggerId}','  
'@{pipeline().TriggerName}','@{pipeline().TriggerTime}',"","","","","","","","","","","","","","","","",""
```

Child pipeline – GetMetadata Activity to get files from Onprem

pl_child_sales_load

Save as template Validate Debug Add trigger

Lookup StartLogging

Get Metadata GetMetadata_Onprem

Filter Filter_Files

General **Dataset** User properties

Dataset * ds_onprem_file_getmetadata Open + New Learn more

Dataset properties

Name	Value
ds_folder_name	@pipeline().parameters.source_folder_...

Filter by last modified

Start time (UTC) End time (UTC)

Field list + New | Delete

- ☐ Argument
- ☐ Child items
- ☐ Item name

Dataset for GetMetadata from onprem

pl_master_sales_load pl_child_sales_load **ds_onprem_file_get...**

01 Binary
ds_onprem_file_getmetadata

Connection **Parameters**

+ New | Delete

☐ Name	Type	Default value
☐ ds_folder_name	String	Value

pl_master_sales_load pl_child_sales_load **ds_onprem_file_get...**

01 Binary
ds_onprem_file_getmetadata

Connection Parameters

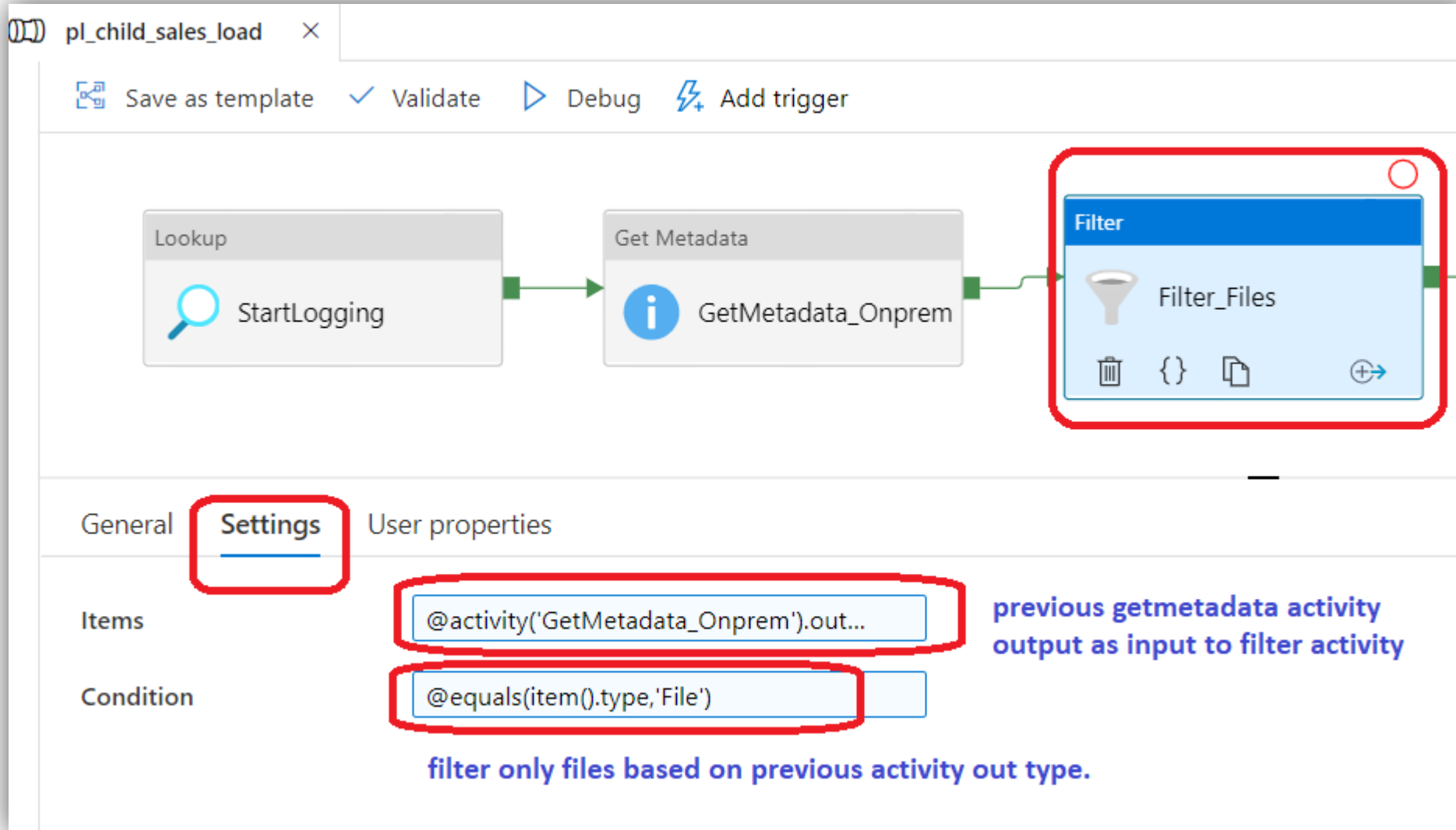
Linked service * ls_file_onprem_source Test connection Edit + New

Integration runtime * OIRSelfhosted (Unavailable) Edit

File path * E:\files\SH / **@dataset().ds_folder_name** / File

Compression type None

Filter-Activity For filter only files.



The screenshot displays the Azure Data Factory (ADF) interface for a workflow named `pl_child_sales_load`. The workflow consists of three activities: `Lookup` (containing `StartLogging`), `Get Metadata` (containing `GetMetadata_Onprem`), and `Filter` (containing `Filter_Files`). The `Filter` activity is highlighted with a red box. Below the workflow, the `Settings` tab is selected, showing the configuration for the `Filter_Files` activity. The `Items` property is set to `@activity('GetMetadata_Onprem').out...`, and the `Condition` is set to `@equals(item().type,'File')`. Both the `Settings` tab and the configuration boxes are highlighted with red boxes. Annotations explain the configuration: the `Items` property is the output of the previous `getmetadata` activity, and the `Condition` filters only files based on the previous activity's output type.

pl_child_sales_load

Save as template Validate Debug Add trigger

Lookup
StartLogging

Get Metadata
GetMetadata_Onprem

Filter
Filter_Files

General **Settings** User properties

Items
`@activity('GetMetadata_Onprem').out...`

Condition
`@equals(item().type,'File')`

previous getmetadata activity output as input to filter activity

filter only files based on previous activity out type.

Foreach Activity

Calling COPY Activity to copy files from on-prem to azure

pl_child_sales_load

Save as template Validate Debug Add trigger

foreach activity to copy files from onprem to azure

Filter

Filter_Files

ForEach

ForEach_CopyFile

Activities

1 activities

Notebook

LandingToStaging

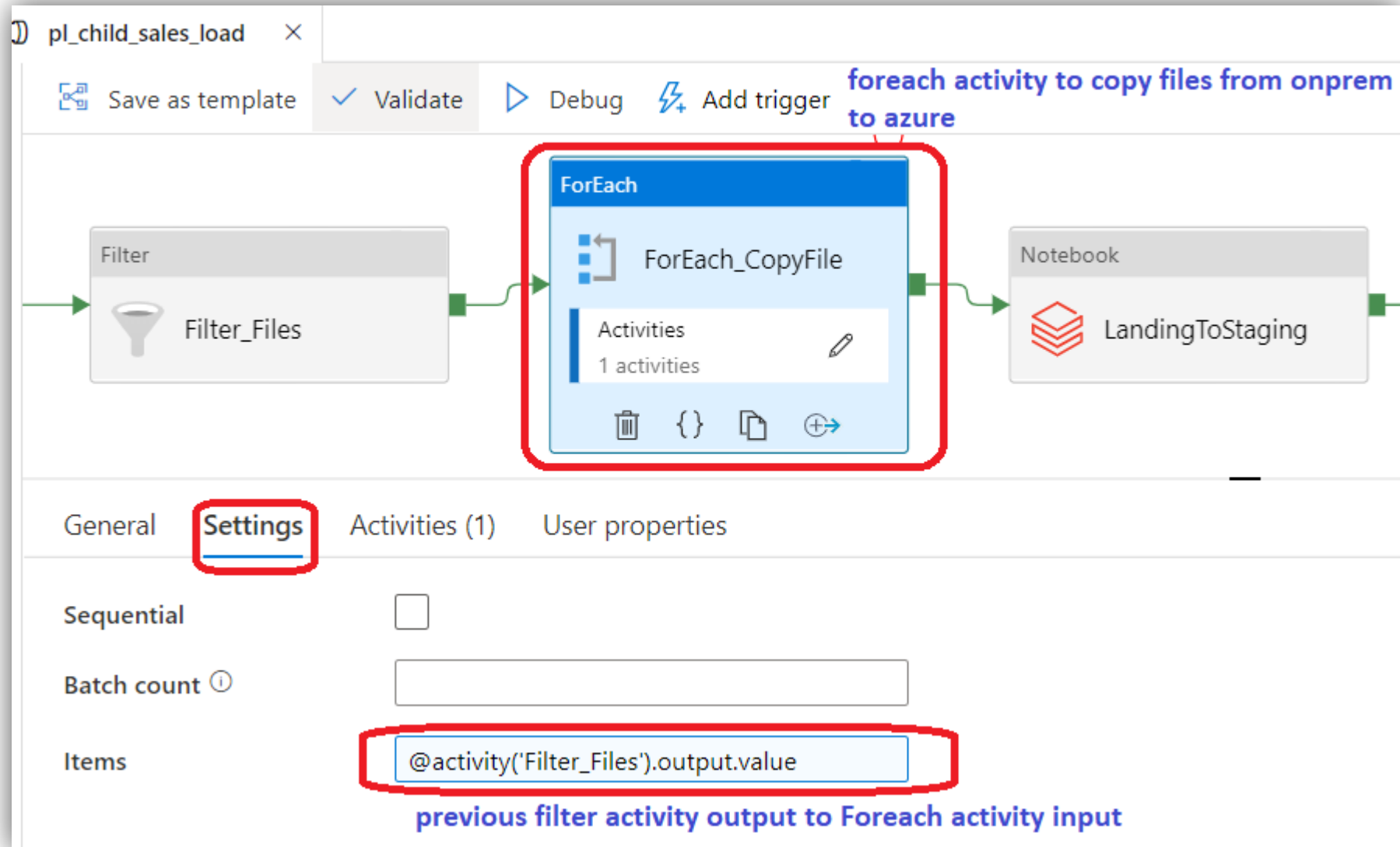
General Settings Activities (1) User properties

Sequential ☐

Batch count

Items

previous filter activity output to Foreach activity input



Copy- Activity

pl_child_sales_load

Save as template Validate Validate copy runtime Debug Add trigger

pl_child_sales_load > ForEach_CopyFile

Copy data

OnPremToLanding

Copy activity for processing files from onprem to azure.
Source: On-prem file
Target: azure Data lake adls gen2

General Source Sink Mapping Settings User properties

Source dataset *

ds_onprem_file_binary

Open + New Learn more

Dataset properties

Name	Value	Type
ds_folder_name	@pipeline().parameters.source_folder_...	string
ds_file_name	@item().name	string

File path type

☒ File path in dataset ☐ File filter ☐ Wildcard file path ☐ List of files

Start time (UTC)End time (UTC)

Filter by last modified

Recursively ☒

Copy-Activity

Source Dataset - Properties

pl_master_sales_load pl_child_sales_load **ds_onprem_file_binary**

01 Binary
ds_onprem_file_binary

Connection **Parameters**

+ New | Delete

☐ Name	Type	Default value
☐ ds_folder_name	String	Value
☐ ds_file_name	String	Value

pl_master_sales_load pl_child_sales_load **ds_onprem_file_binary**

01 Binary
ds_onprem_file_binary

Connection Parameters

Linked service * ls_file_onprem_source Test connection Edit + New Learn more

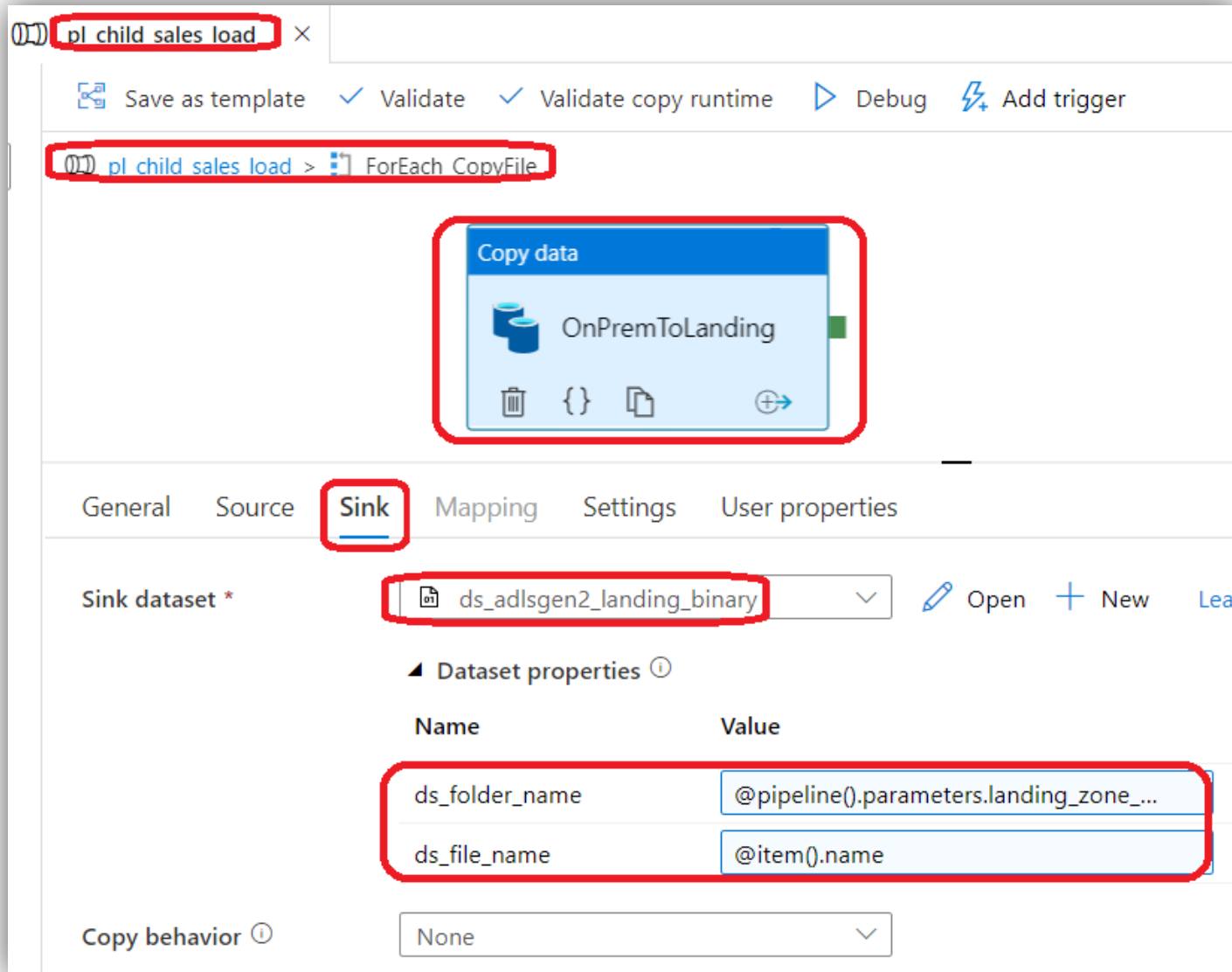
Integration runtime * OIRSelfhosted (Unavailable) Edit

File path * E:\files\SH / @dataset().ds_folder_name / @dataset().ds_file_name

Compression type None

Copy-Activity

Sink- Dataset properties



The screenshot shows the configuration of a Copy Activity in Azure Data Factory. The activity is named "pl child sales load". The Sink dataset is configured with the following properties:

Sink dataset *: ds_adlsgen2_landing_binary

Dataset properties

Name	Value
ds_folder_name	@pipeline().parameters.landing_zone_...
ds_file_name	@item().name

Copy behavior: None

Sink DataSet Parameters and dynamic filepath.

pl_master_sales_load pl_child_sales_load ds_adlsgen2_landing... X

01 Binary
ds_adlsgen2_landing_binary

Connection **Parameters**

+ New | Delete

☐ Name	Type	Default value
☐ ds_folder_name	String	Value
☐ ds_file_name	String	Value

pl_master_sales_load pl_child_sales_load ds_adlsgen2_landing... X

01 Binary
ds_adlsgen2_landing_binary

Connection Parameters

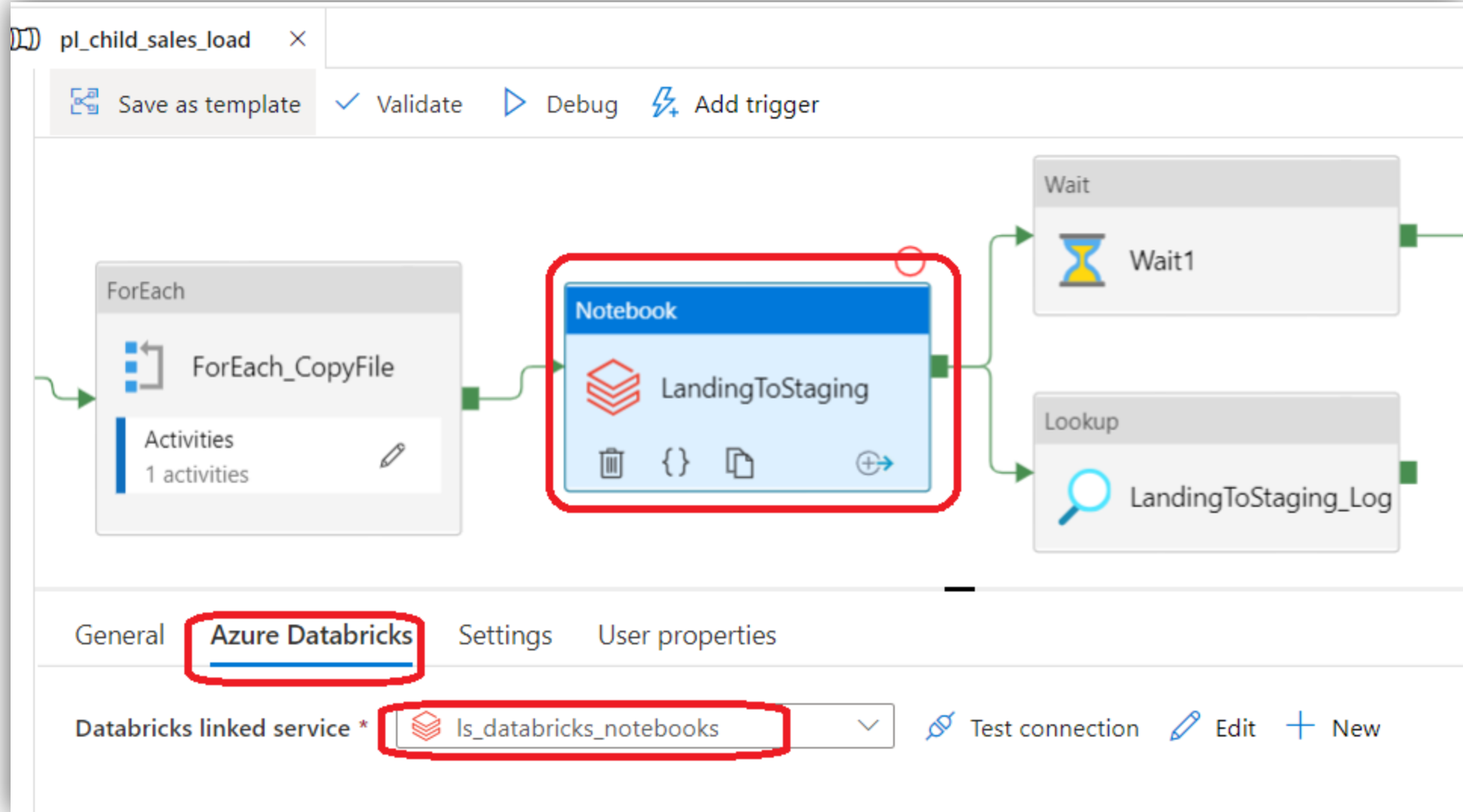
Linked service * Is_adlsgen2_sales Test connection Edit + New Learn more

File path * sales / @concat('landing/sales/',dataset().ds_f... / @dataset().ds_file_name

Compression type None

@concat('landing/sales/',dataset().ds_folder_name)

Calling Databricks Notebook activity for Landing to Staging.



Notebook : Landing to staging widgets parameters

pl_child_sales_load

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

ForEach

ForEach_CopyFile

Activities
1 activities

Notebook

LandingToStaging

Wait

Wait1

Lookup

LandingToStaging_Log

General Azure Databricks **Settings** User properties

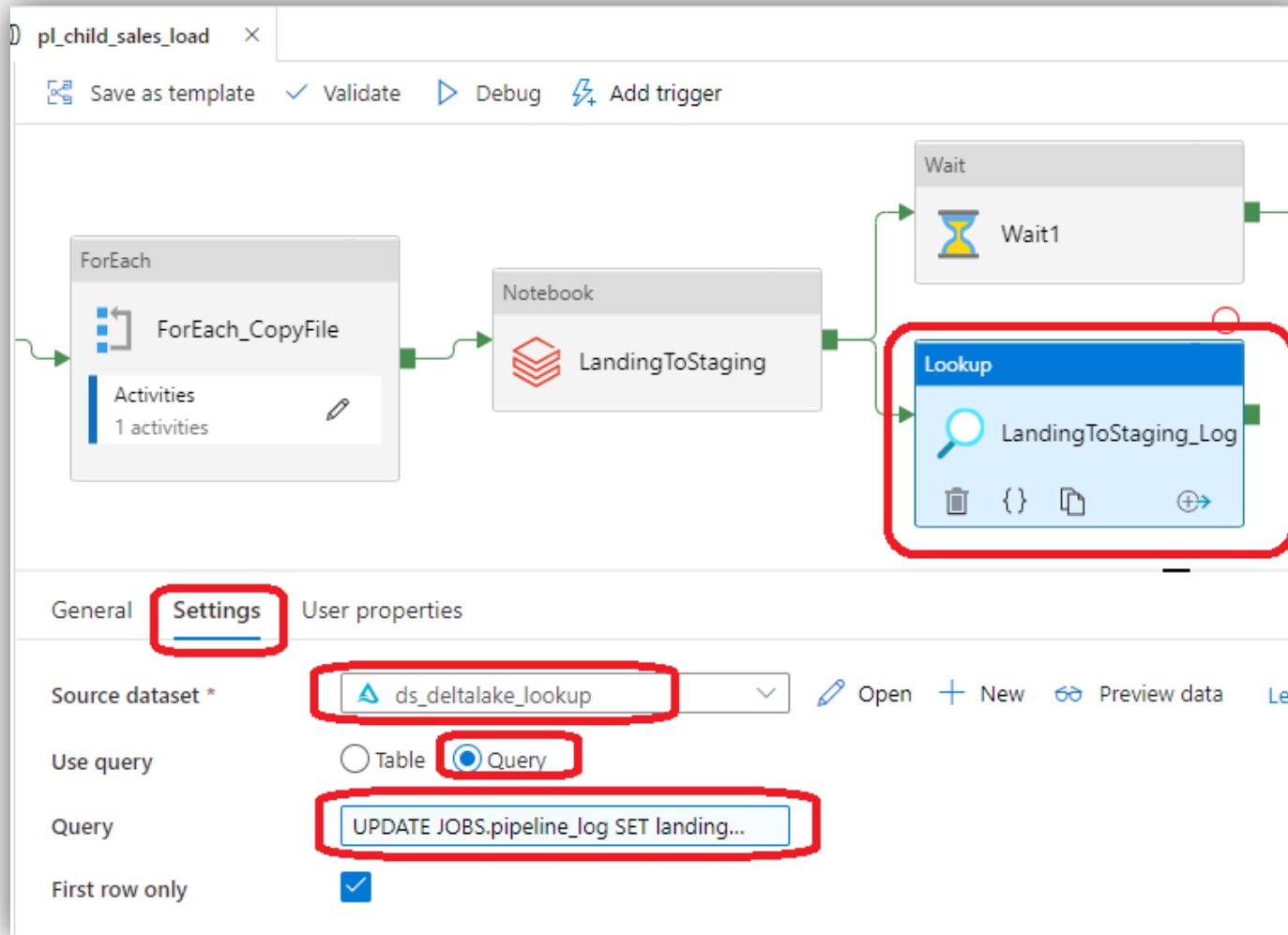
Notebook path * Browse Open

Base parameters

+ New | Delete

☐ Name	Value
☐ landing_zone_file_pat	@pipeline().parameters.landing_zone_...
☐ landing_zone_folder_r	@pipeline().parameters.landing_zone_...
☐ landing_zone_file_nam	@pipeline().parameters.landing_zone_...
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...
☐ pyspark_schema	@pipeline().parameters.pyspark_schema
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id

Lookup- Activity for logging into delta lake table



The screenshot shows the configuration of a Lookup activity named 'LandingToStaging_Log' within a pipeline named 'pl_child_sales_load'. The pipeline flow is as follows:

- ForEach** activity (containing 'ForEach_CopyFile') connects to the **Notebook** activity 'LandingToStaging'.
- The **Notebook** activity connects to the **Lookup** activity 'LandingToStaging_Log'.
- The **Lookup** activity connects to a **Wait** activity 'Wait1'.

The **Lookup** activity configuration is shown in the bottom panel:

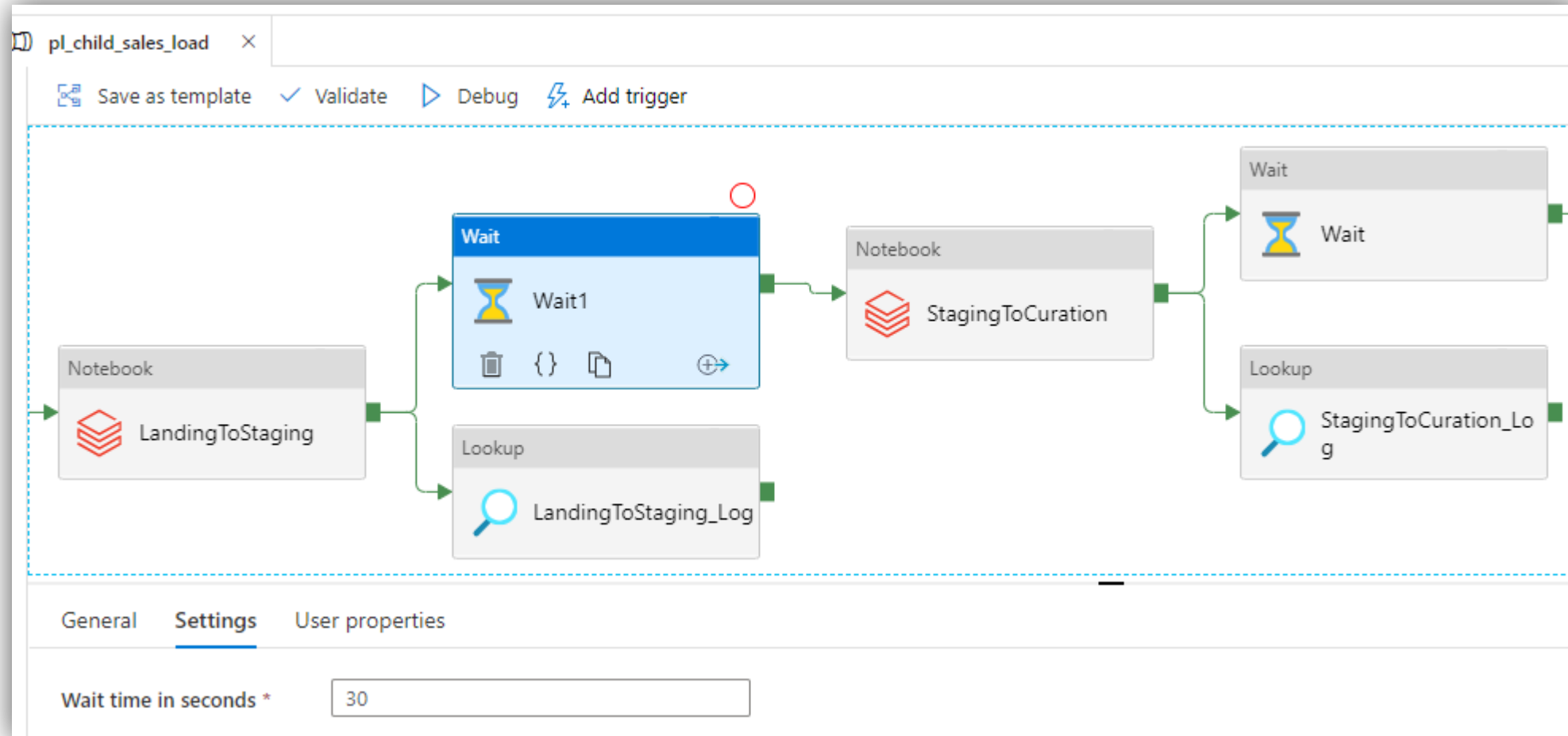
- Source dataset ***: ds_deltalake_lookup
- Use query**: ☒ Query
- Query**: UPDATE JOBS.pipeline_log SET landing...
- First row only**: ☒

UPDATE JOBS.pipeline_log SET

landing_rows=@{activity('LandingToStaging').output.runOutput.landingRows},rejected_rows=@{activity('LandingToStaging').output.runOutput.rejectedRows},
staging_rows=@{activity('LandingToStaging').output.runOutput.stagingRows} WHERE pipeline_runid='@{pipeline().RunId}'

Wait- Activity

Wait activity to wait job execution few seconds (30 sec) before staging next notebook



Notebook : Activity for Staging to Curation

pl_child_sales_load

Save as template Validate Debug Add trigger

Wait1

Lookup LandingToStaging_Log

Notebook StagingToCuration

Wait

Lookup StagingToCuration_Log

General Azure Databricks **Settings** User properties

Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sale: Browse Open

Base parameters

+ New | Delete

Name	Value
staging_zone_databas	@pipeline().parameters.staging_zone_...
staging_zone_table_n:	@pipeline().parameters.staging_zone_t...
staging_zone_table_pl	@pipeline().parameters.staging_zone_t...
job_name	@pipeline().parameters.job_name
job_id	@pipeline().parameters.job_id
curation_zone_databa	@pipeline().parameters.curation_zone...
curation_zone_table_n	@pipeline().parameters.curation_zone...
curation_zone_table_p	@pipeline().parameters.curation_zone...

Lookup- Activity for updating log

The screenshot displays the Azure Data Factory pipeline editor for a pipeline named 'pl_child_sales_load'. The pipeline flow includes a 'Wait' activity (Wait1), a 'Lookup' activity (LandingToStaging_Log), a 'Notebook' activity (StagingToCuration), and another 'Wait' activity. A 'Lookup' activity (StagingToCuration_Log) is highlighted with a red box. Below the pipeline canvas, the 'Settings' tab is selected, showing the configuration for the 'StagingToCuration_Log' activity. The 'Source dataset' is set to 'ds_deltalake_lookup'. The 'Use query' option is selected, and the 'Query' field contains the SQL statement: 'UPDATE JOBS.pipeline_log SET curatio...'. The 'First row only' checkbox is checked.

General **Settings** User properties

Source dataset * ds_deltalake_lookup Open New Preview data Learn more

Use query ☐ Table ☒ Query

Query UPDATE JOBS.pipeline_log SET curatio...

First row only ☒

UPDATE JOBS.pipeline_log SET
curation_read_rows=@{activity('StagingToCuration').output.runOutput.curationRows['num_affected_rows']},curation_inserted_rows=@{activity('StagingToCuration').output.runOutput.curat
ionRows['num_inserted_rows']},curation_updated_rows=@{activity('StagingToCuration').output.runOutput.curationRows['num_updated_rows']} WHERE
pipeline_runid='@{pipeline().RunId}'

IF-Condition : for processing Dimensions based on condition (table_type='DIM')

pl_child_sales_load

Save as template Validate Debug Add trigger

Wait

Lookup

StagingToCuration_Loading

If Condition

IFDim

True

2 activities

False

No activities

Switch

SwitchFacts

Default

No activities

sales_transaction

1 activities

costs_transaction

1 activities

General Activities (2) User properties

Expression

@equals(pipeline().parameters.table_t...

@equals(pipeline().parameters.table_type,'dim')

Case	Activity
True	2 Activities
False	No activities

Notebook: Curation To DWH processing inside if condition

pl_child_sales_load

Save as template Validate Debug Add trigger

pl_child_sales_load > IFDim > True activities

Notebook

CurationToDWHDim

Lookup

CurationToDWHDim_Log

General Azure Databricks **Settings** User properties

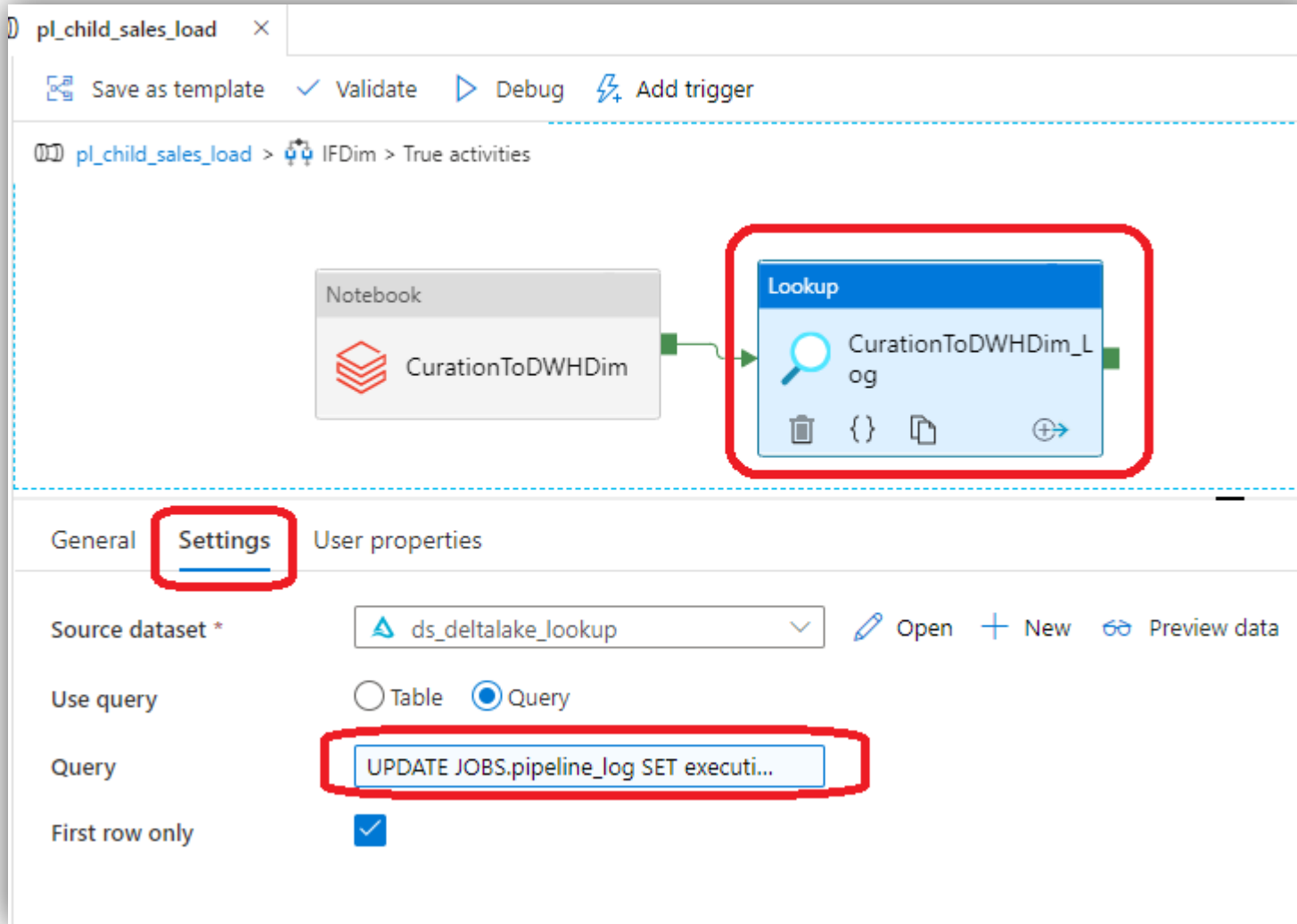
Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sale: Browse Open

▲ Base parameters

+ New | Delete

☐	Name	Value
☐	job_name	@pipeline().parameters.job_name
☐	job_id	@pipeline().parameters.job_id
☐	curation_zone_databa	@pipeline().parameters.curation_zone...
☐	curation_zone_table_n	@pipeline().parameters.curation_zone...
☐	curation_zone_table_p	@pipeline().parameters.curation_zone...
☐	dw_zone_database_na	@pipeline().parameters.dw_zone_data...
☐	dw_zone_table_name	@pipeline().parameters.dw_zone_table...
☐	dw_zone_table_pk_col	@pipeline().parameters.dw_zone_table...

Updating Log in log table



The screenshot shows the Databricks IDE interface for a pipeline named `pl_child_sales_load`. The pipeline is currently in the `IFDim` state, and the `True` activities are visible. A `Notebook` activity named `CurationToDWHDim` is connected to a `Lookup` activity. The `Lookup` activity is highlighted with a red box and contains the following configuration:

- Source dataset:** `ds_deltalake_lookup`
- Use query:** ☒ Query
- Query:** `UPDATE JOBS.pipeline_log SET executi...` (highlighted with a red box)
- First row only:** ☒

```
UPDATE JOBS.pipeline_log SET
execution_status='SUCCESS',dwh_read_rows=@{activity('CurationToDWHDim').output.runOutput.dwhRows['num_affected_rows']],dwh_inserted_rows=@{activity('CurationToDWHDim').output.runOutput.dwhRows['num_inserted_rows']],dwh_updated_rows=@{activity('CurationToDWHDim').output.runOutput.dwhRows['num_updated_rows']}] WHERE pipeline_runid=@{pipeline().RunId}'
```

Switch – Case : for processing Fact data loads based on condition.

pl_child_sales_load

Save as template Validate Debug Add trigger

Wait

Lookup

StagingToCuration_Loading

If Condition

IFDim

True

2 activities

False

No activities

Switch

SwitchFacts

Default

No activities

sales_transaction

1 activities

costs_transaction

1 activities

General Activities (2) User properties

Expression

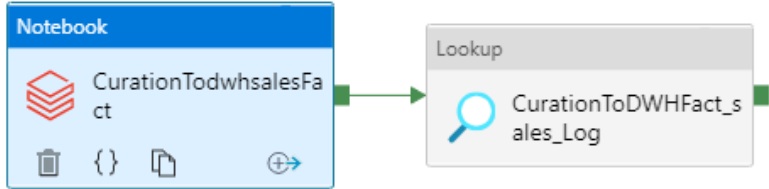
@pipeline().parameters.job_name

+ Add case

Case	Activity
Default	No activities
sales_transaction	1 Activity
costs_transaction	1 Activity

Processing both Facts inside Switch case

pl_child_sales_load > SwitchFacts - salestransaction



General Azure Databricks **Settings** User properties

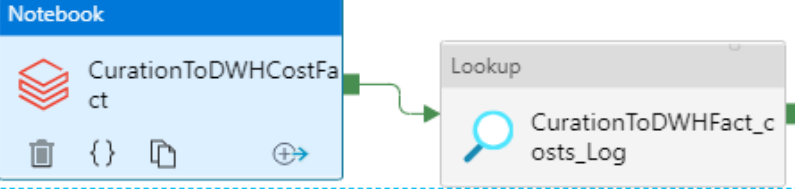
Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sale: Browse Open

▲ Base parameters

+ New | Delete

☐ Name	Value
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id
☐ curation_zone_databa	@pipeline().parameters.curation_zone...
☐ curation_zone_table_n	@pipeline().parameters.curation_zone...
☐ curation_zone_table_p	@pipeline().parameters.curation_zone...
☐ dw_zone_database_na	@pipeline().parameters.dw_zone_data...
☐ dw_zone_table_name	@pipeline().parameters.dw_zone_table...
☐ dw_zone_table_pk_col	@pipeline().parameters.dw_zone_table...
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n:	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...

pl_child_sales_load > SwitchFacts - coststransaction



General Azure Databricks **Settings** User properties

Notebook path * /Shared/Pyspark_Project_Dynamic/NB_cost: Browse Open

▲ Base parameters

+ New | Delete

☐ Name	Value
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id
☐ curation_zone_databa	@pipeline().parameters.curation_zone...
☐ curation_zone_table_n	@pipeline().parameters.curation_zone...
☐ curation_zone_table_p	@pipeline().parameters.curation_zone...
☐ dw_zone_database_na	@pipeline().parameters.dw_zone_data...
☐ dw_zone_table_name	@pipeline().parameters.dw_zone_table...
☐ dw_zone_table_pk_col	@pipeline().parameters.dw_zone_table...
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n:	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...

All The Best 😊